	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.	**Assessment criteria:** C8: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. C9: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C8 and C9. Answer individually on separate paper. Questions 1–2 assess C8; each question is worth 3 points: 1 point for task (a) and 2 points for task (b). Questions 3–5 assess C9; each problem is worth 2 points: 1 point for the filtering tree and interpretation and 1 point for the supporting table. Each criterion totals 6 points. A **minimum of 4 points** is required to pass each criterion. Partial credit is available for correct comparisons, filtering stages, and justified reasoning even if later conclusions contain errors. The final C10 section is an unscored reminder of the separate activity.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	6	6

1. Repair Workshop Plans [C8-3]

A bicycle repair workshop chooses next quarter's capacity before demand is known. Profits are in thousand USD; forecast probabilities are shown below.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	-20
Flexible (A_2)	85	65	10
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

Decision statistics (Regret means maximum regret):

Alternative	Max	Min	Regret	EMV
A_1	120	-20	55	35
A_2	85	10	35	58
A_3	50	35	70	44
A_4	75	20	50	63

(a) Report each criterion's recommended alternative(s), deciding statistic, and value. **[1]**

(b) A subsidy s (integer, $0 \le s \le 30$, thousand USD) increases the low-demand payoffs of both A_1 and A_2 by s. All other payoffs and probabilities stay fixed. An alternative is eligible only if its profit is nonnegative in every state. Determine the ranges of s for which each alternative is eligible. Find the smallest s that changes the Maximin recommendation from part (a) to A_2 alone. At that s, report all four recommendations and deciding values among eligible alternatives. **[2]**

2. Community Market Stall [C8-3]

An organizer chooses a weekend market stall. Profits are in hundred USD; forecast probabilities are shown below.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	48	32

Alternative	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	48	32	32	38.4

(a) Report each criterion's recommended alternative(s), deciding statistic, and value. **[1]**

(b) A sponsor guarantees C_2 a high-attendance profit of g, replacing 48, where g is an integer, $48 \le g \le 80$, in hundred USD. All other payoffs and probabilities stay fixed; both stalls remain eligible. Find the smallest g that changes the Minimax Regret recommendation from part (a) to C_2 alone. Justify the threshold and boundary tie; interpret the increase from 48 in USD. At that g, report all four updated recommendations and deciding values, including every tie. **[2]**

3. Book Fair Stock [C9-2]

A bookseller chooses a fair stock plan D_1–D_6. Net profits are in hundred USD under high, normal, and low sales, with probabilities 0.5, 0.3, and 0.2.

Alternative	Max	Min	Regret	EMV
D_1	100	0	40	65
D_2	90	10	30	65
D_3	80	20	20	65
D_4	70	30	30	59
D_5	60	40	40	53
D_6	50	30	50	43

Apply these stages in order:

1. EMV/Bayes: keep the 2 highest EMVs, retaining cutoff ties.
2. Maximin: select the highest Min among survivors.

Complete the tree, table, and interpretation described above.

4. Refill Shop Expansion [C9-2]

A refill shop chooses next year's expansion plan F_1–F_6. Net profits are in thousand USD under strong, steady, and weak demand, with probabilities 0.3, 0.5, and 0.2.

Alternative	Max	Min	Regret	EMV
F_1	120	0	45	61
F_2	105	15	30	64.5
F_3	95	25	25	68.5
F_4	85	35	35	70
F_5	75	45	45	64
F_6	65	40	55	55

Apply these stages in order:

1. Minimax Regret: keep the 5 lowest Regret values.
2. EMV/Bayes: keep the 4 highest EMVs among survivors.
3. Maximax: keep the 2 highest Max values among survivors.
4. Maximin: select the highest Min among survivors.

Construct the filtering tree and supporting table; include rules, values, eliminations, final choice, and a one-sentence interpretation. Use the shared rules for Questions 3–5.

5. School Supply Orders [C9-2]

A school supplier chooses next term's order plan H_1–H_6. Net profits are in hundred USD under high, normal, and low demand, with probabilities 0.3, 0.5, and 0.2.

Alternative	Max	Min	Regret	EMV
H_1	120	0	45	61
H_2	105	15	30	64.5
H_3	95	25	25	68.5
H_4	85	35	35	70
H_5	75	45	45	64
H_6	65	40	55	55

Apply these stages in order:

1. Maximin: keep the 4 highest Min values.
2. Minimax Regret: keep the 3 lowest Regret values among survivors.
3. Maximax: keep the 2 highest Max values among survivors.
4. EMV/Bayes: select the highest EMV among survivors.

Complete the tree, table, and interpretation described above. Use the shared rules for Questions 3–5.

C10: Separate activity reminder (unscored)

C10 is assessed through the separate **Interactive Decision Webpage: C10 Practice** activity.

Complete the individual interactive decision webpage described in that activity and submit the **deployed webpage URL**. Refer to that activity for the full instructions. C10 has no exam questions or points.